Four Fundamental Subspaces

Example 1:

$$A = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 5 \end{bmatrix}$$

A

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 5 \end{bmatrix} \quad r_3 = r_3 - r_1$$

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{bmatrix} \quad r_3 = r_3 - r_2$$

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad r_1 = r_1 - r_2$$

$$\begin{array}{c} \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ \begin{array}{cccc} p & p & f & f \end{array} \end{array}$$

RREF(A)

A^T

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 5 \end{bmatrix} \quad \begin{array}{l} r_2 = r_2 - r_1 \\ r_3 = r_3 - r_1 \\ r_4 = r_4 - 2r_1 \end{array}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 3 & 3 \end{bmatrix} \quad \begin{array}{l} r_3 = r_3 - 2r_2 \\ r_4 = r_4 - 3r_2 \end{array}$$

$$\begin{array}{c} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \begin{array}{ccc} p & p & f \end{array} \end{array}$$

RREF (A^T)

$$RREF(A) = \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad RREF(A^T) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Properties	Columnspace	Rowspace	Nullspace	Left Nullspace
	$C(A)$	$C(A^T)$	$N(A)$	$N(A^T)$
dimension	$r = 2$	$r = 2$	$n - r = 4 - 2$	$m - r = 3 - 1$
Bases	$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 0 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$
Subspace in	$\mathbb{R}^3$	$\mathbb{R}^4$	$\mathbb{R}^4$	$\mathbb{R}^3$

$$\dim(C(A^T)) + \dim(N(A))$$
$$= 2 + 2 = 4$$
$$\mathbb{R}^4$$

$$\dim(C(A)) + \dim(N(A^T))$$
$$2 + 1 = 3$$
$$\mathbb{R}^3$$